YUXUAN WANG

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EDUCATION

Nanyang Technological University (NTU), Computer Science and Engineering,
Doctor of Philosophy — Aug. 2022 - Apr. 2026

Supervised by Prof. *Hanwang Zhang* in Computer Vision.
Thesis Title: *Towards Scalable 3D Generation and Consistent 3D Manipulation*

National University of Singapore (NUS), Electrical and Computer Engineering,
Master of Science (*GPA: 4.47/5*) — Aug. 2021 - Jun. 2022

Supervised by Prof. *Mike Zheng Shou* in Computer Vision.

Beihang University, Electronic Information Engineering,
Bachelor of Engineering (*GPA: 87.5/100*) — Sep. 2016 - Jun. 2020

EXPERIENCE

Research Scientist | ByteDance Seed
Seed 3D — *Current*

Research Intern | Tencent Hunyuan
Hunyuan 3D — Jun. 2024

FIRST-AUTHOR PUBLICATIONS

Personalize Your Gaussian: Consistent 3D Scene Personalization from a Single Image
Yuxuan Wang, Xuanyu Yi, Qingshan Xu, Yuan Zhou, Long Chen, Hanwang Zhang
AAAI Conference on Artificial Intelligence (AAAI), 2026

- We present Consistent Personalization for 3D Gaussian Splatting (CP-GS), a framework that progressively propagates the single-view reference appearance to novel perspectives, offering high-quality 3DGS personalization with faithful referential alignment.

Nautilus: Locality-aware Autoencoder for Scalable Mesh Generation
Yuxuan Wang*, Xuanyu Yi*, Haohan Weng*, Xiaokang Wei, Xianghui Yang, Chunchao Guo, Long Chen, Hanwang Zhang
International Conference on Computer Vision (ICCV), 2025

- We propose Nautilus, a locality-aware autoencoder for artist-like mesh generation, which leverages the local properties of manifold meshes to achieve structural fidelity and efficient representation.

View-Consistent 3D Editing with Gaussian Splatting
Yuxuan Wang, Xuanyu Yi, Zike Wu, Na Zhao, Long Chen, Hanwang Zhang
European Conference on Computer Vision (ECCV), 2024

- In the diffusion model, we proposed effective multi-view consistency designs that harmonize the inconsistent multi-view image guidance by integrating with 3D Gaussian Splatting (3DGS) characteristics, offering high-quality 3DGS editing.

Predicate Debiasing in Vision-Language Models Integration for Scene Graph Generation
Yuxuan Wang, Xiaoyuan Liu
Main Conference, Empirical Methods in Natural Language Processing (EMNLP), 2024

- We introduced a plug-and-play debiasing method for the zero-shot VLMs, dynamically ensembling them to address the underrepresentation issue in Scene Graph Generation (SGG) models.

GEB+: A Benchmark for Generic Event Boundary Captioning, Grounding and Retrieval
Yuxuan Wang, Difei Gao, Licheng Yu, Stan Weixian Lei, Matt Feiszli, Mike Zheng Shou
European Conference on Computer Vision (ECCV), 2022

- We introduced three tasks of video boundary understanding on our new dataset called Kinetics-GEB+ (Generic Event Boundary Plus), consisting of over 170k boundaries associated with captions in 12K videos.
- We designed a new Temporal-based Pairwise Difference (TPD) Modeling method for visual difference representation and achieved significant performance improvements.

CO-AUTHOR PUBLICATIONS

Pushing Rendering Boundaries: Hard Gaussian Splatting
Qingshan Xu, Jiequan Cui, Xuanyu Yi, Yuxuan Wang, Yuan Zhou, Yew-Soon Ong, Hanwang Zhang
AAAI Conference on Artificial Intelligence (AAAI), 2026

NeuSpring: Neural Spring Fields for Reconstruction and Simulation of Deformable Objects from Videos
Qingshan Xu, Jiao Liu, Shangshu Yu, Yuxuan Wang, Yuan Zhou, Junbao Zhou, Jiequan Cui, Yew-Soon Ong, Hanwang Zhang
AAAI Conference on Artificial Intelligence (AAAI), 2026

PBR3DGen: A VLM-guided mesh generation with high-quality PBR texture
Xiaokang Wei, Bowen Zhang, Xianghui Yang, Yuxuan Wang, Xi Zhao, Yan Luximon
AAAI Conference on Artificial Intelligence (AAAI), 2026

DragNeXt: Rethinking Drag-Based Image Editing
Yuan Zhou, Junbao Zhou, Qingshan Xu, Kesen Zhao, Yuxuan Wang, Hao Fei, Richang Hong, Hanwang Zhang
AAAI Conference on Artificial Intelligence (AAAI), 2026

Symbolic Replay: Scene Graph as Prompt for Continual Learning on VQA Task
Stan Weixian Lei, Difei Gao, Jay Zhangjie Wu, Yuxuan Wang, Wei Liu, Mengmi Zhang, Mike Zheng Shou
AAAI Conference on Artificial Intelligence (AAAI), 2023, **Oral**

AssistSR: Task-oriented Video Segment Retrieval for Personal AI Assistant
Stan Weixian Lei, Difei Gao, Yuxuan Wang, Dongxing Mao, Zihan Liang, Lingmin Ran, Mike Zheng Shou
Findings, Empirical Methods in Natural Language Processing (EMNLP), 2022

RESEARCH PROJECTS

IMS-MLD Decoding Algorithm for Reed-Muller Code
Senior Thesis Supervised by Prof. *Qin Huang*, Beihang University Feb. 2020

Blind Identification and Demodulation of Modulated Signals
Supervised by Prof. *Qin Huang*, Beihang University Jul. 2018

SKILLS

- **Language**: Mandarin (Native), English (GRE: 331/340, TOEFL: 110/120)
- **Coding**: Python, Java, C, MATLAB, SQL
- **Music**: Piano, Guitar, Clarinet, Cavalry trumpet, Singing, Chorus Conducting